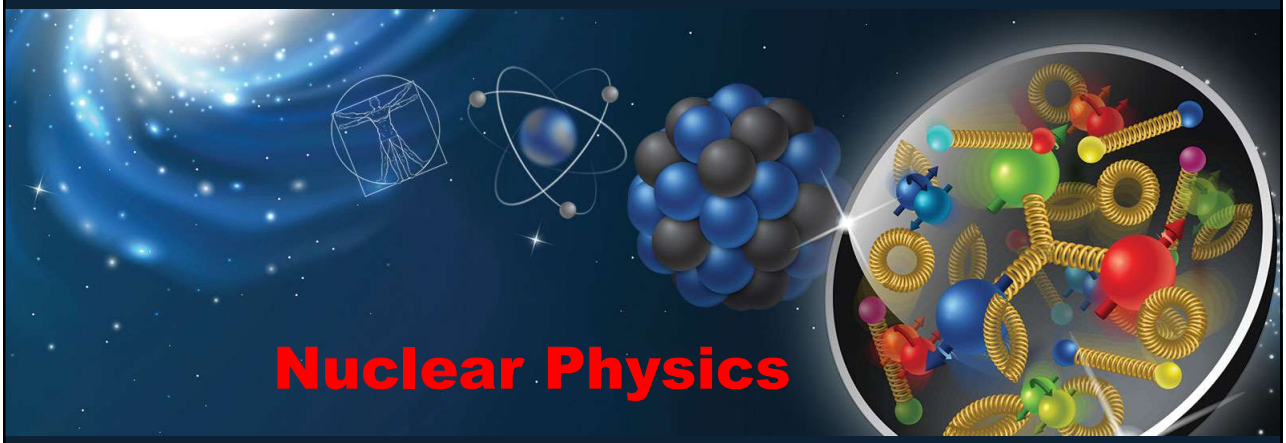
PHY1203 - Modern Physics
2025 - Lesson 08



Nuclear Physics

Nuclear Reactions

It is possible to change nuclei by bombarding with energetic nuclear particles. **Rutherford** was the first to observe nuclear reactions by bombarding alpha particles (from natural radioactive sources) with nitrogen (N) atoms.

He found the **production of protons**.

$$^{14}_{7}\mathrm{N} + ^{4}_{2}\mathrm{He} \rightarrow ^{17}_{8}\mathrm{C} + ^{1}_{1}\mathrm{H}$$

In 1932 Sir James **Chadwick** in England bombarded a beryllium target with alpha particles and identify the production of carbon-12.

$$^{9}_{4}\mathrm{Be} + ^{4}_{2}\mathrm{He} \rightarrow ^{12}_{6}\mathrm{C} + ^{A}_{Z}\mathrm{X}$$

Identify the product $^{A}_{Z}\mathrm{X}$. Answer: $^{1}_{0}\mathrm{n}$. This was identified as the neutron

This was the experiment Chadwick used to discover neutron !

The Q Value

The Q-value of a nuclear reaction is the energy released or absorbed during the reaction.

It is calculated from the mass difference between reactants and products using the Einstein's mass-energy equivalence ($\Delta E = \Delta m.c^2$).

- A positive (+) Q-value indicates that energy is released (**exothermic reaction**)
- A negative (-) Q-value means energy must be provided for the reaction to occur (**endothermic reaction**).

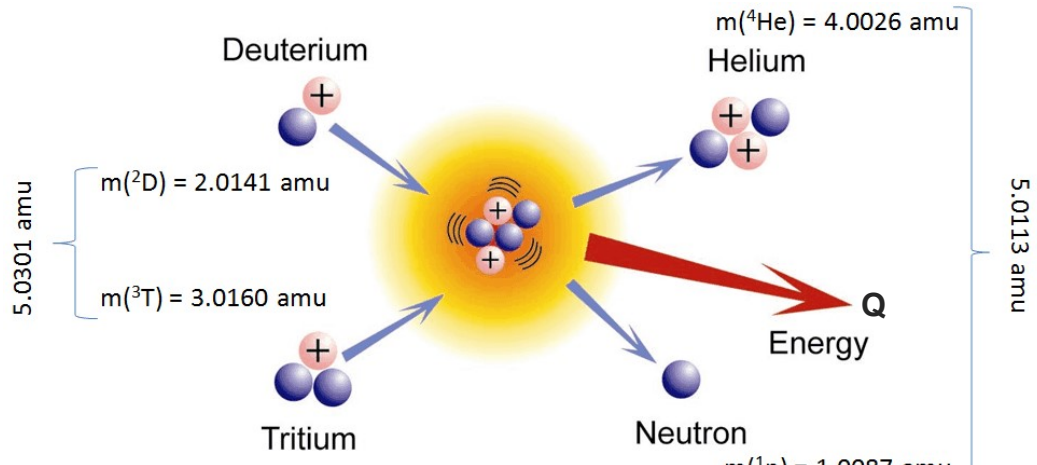
e.g.(1) Find the Q-value of the reaction

$$^{14}_{7}\mathrm{N} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He} + \mathrm{Q}$$

e.g.(1) Find the Q-value of the reaction $^{14}_{7}\mathrm{N} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He} + \mathrm{Q}$

$$^{14}_{7}\mathrm{N} = 14.003\,074\ \mathrm{u} \qquad ^{12}_{6}\mathrm{C} = 12.000\,000\ \mathrm{u}$$
$$^{2}_{1}\mathrm{H} = 2.014\,102\ \mathrm{u} \qquad ^{4}_{2}\mathrm{He} = 4.002\,602\ \mathrm{u}$$
$$= 16.017\,176\ \mathrm{u} \qquad = 16.002\,602\ \mathrm{u}$$

$$\Delta m = 16.017\,176\ \mathrm{u} - 16.002\,602\ \mathrm{u} = 0.014\,574\ \mathrm{u}$$
$$Q = 0.014\,574 \times 931.5\ \mathrm{MeV}$$
$$Q = +13.576\ \mathrm{MeV}$$ (Exothermic reaction)



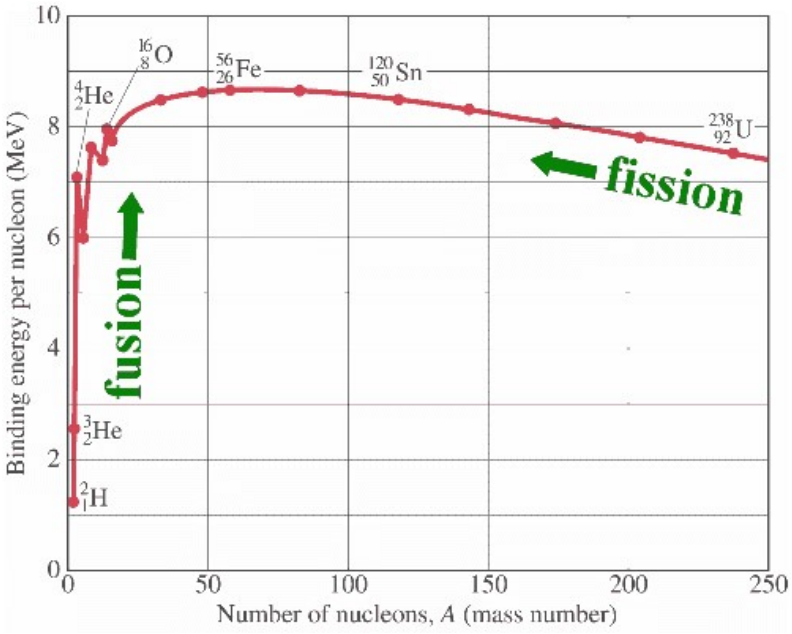
Q = 0.0188 amu x 931.481 MeV/amu = 17.5 MeV

e.g.(2) Find the Q-value of the reaction $^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H} + \mathrm{Q}$

$^{14}_{7}\mathrm{N} = 14.003\,074\ \mathrm{u}$
$^{4}_{2}\mathrm{He} = \ 4.002\,602\ \mathrm{u}$
$= 18.005\,676\ \mathrm{u}$

$^{17}_{8}\mathrm{O} \ = 16.999\,133\ \mathrm{u}$
$^{1}_{1}\mathrm{H} = \ 1.007\,825\ \mathrm{u}$
$= \ 18.006\,958\ \mathrm{u}$

$\Delta m = 18.005\,676\ \mathrm{u} - \ 18.006\,958\ \mathrm{u} = -0.001\,282\ \mathrm{u}$
$Q \ = -0.001\,282 \ \times 931.5\ \mathrm{MeV}$
$Q \ = -1.194\ \mathrm{MeV}$ (Endothermic reaction)



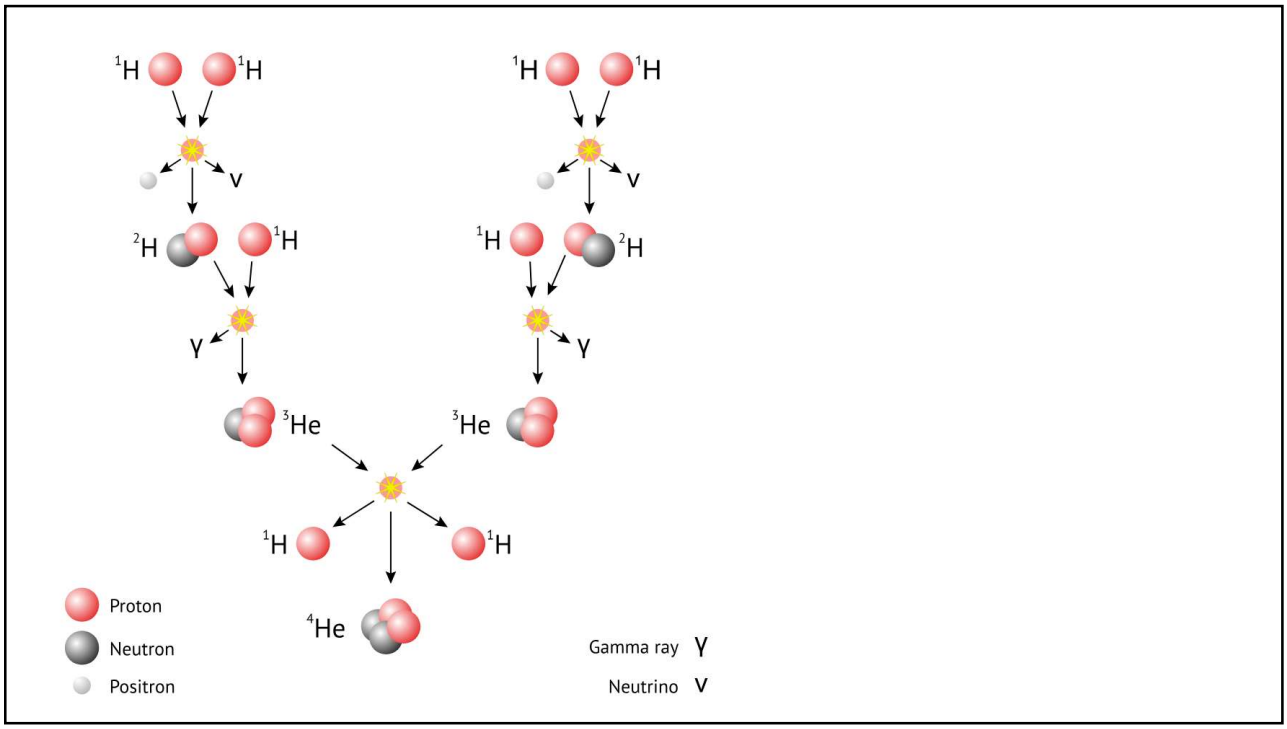
(1) Fusion reaction

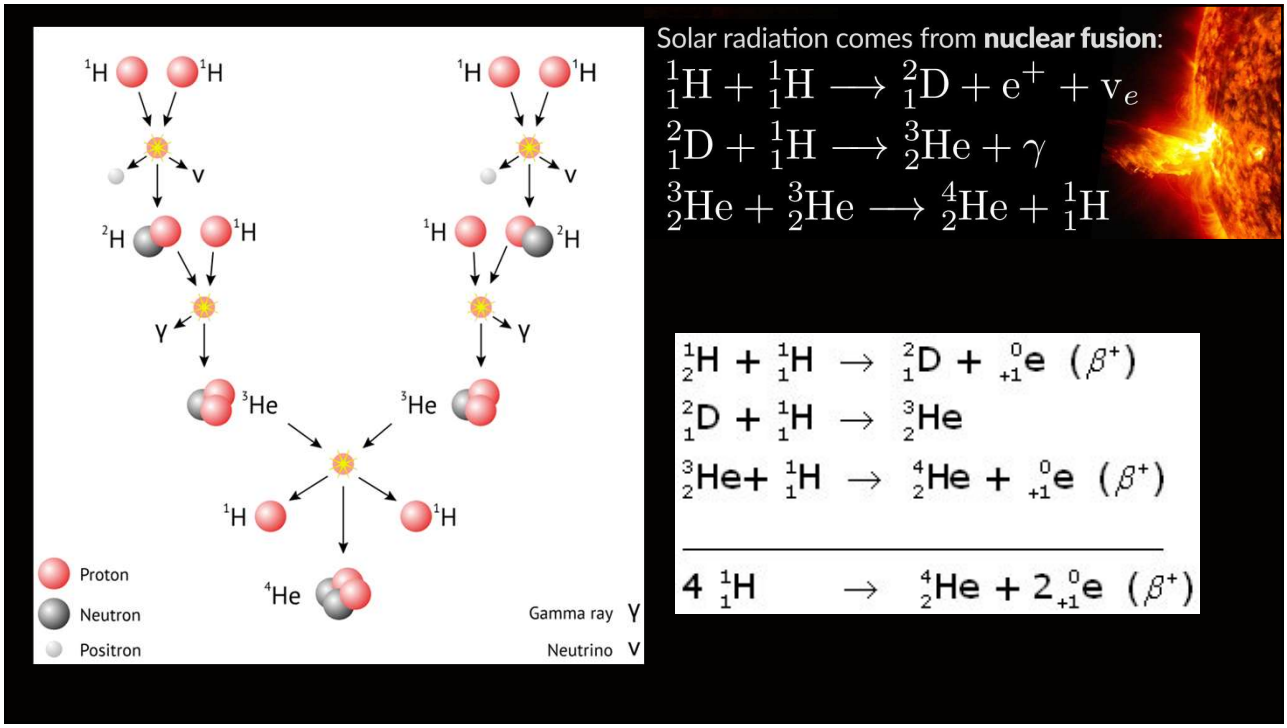
A nuclear fusion reaction is a process where two light atomic nuclei combine to form a single, heavier nucleus, releasing a massive amount of energy.

The fusion reaction in the sun and stars, occur under very high temperature and extreme pressure that force nuclei to overcome electrostatic repulsion and fuse via the strong nuclear force.

The Sun's core is hot enough (15,000,000 °C) to turn hydrogen gas into a plasma,

The huge gravity of the Sun creates the extreme pressure needed to confine the hydrogen nuclei in a small space,


^{1}H ^{1}H
ν
^{2}H ^{1}H
γ
^{3}He
^{1}H ^{1}H
^{4}He
^{1}H ^{1}H
ν
^{1}H ^{2}H
γ
^{3}He
Proton
Neutron
Positron
Gamma ray γ
Neutrino ν

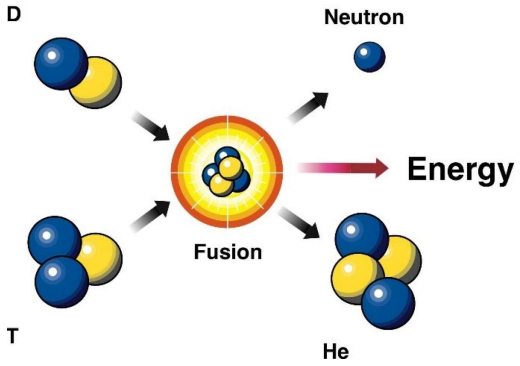

^{1}H ^{1}H
ν
^{2}H ^{1}H
γ
^{3}He
^{1}H ^{1}H
^{4}He
^{1}H ^{1}H
ν
^{1}H ^{2}H
γ
^{3}He
Proton
Neutron
Positron
Gamma ray γ
Neutrino ν
Solar radiation comes from **nuclear fusion**:
$^1_1\mathrm{H} + {}^1_1\mathrm{H} \longrightarrow {}^2_1\mathrm{D} + \mathrm{e}^+ + \nu_e$
$^2_1\mathrm{D} + {}^1_1\mathrm{H} \longrightarrow {}^3_2\mathrm{He} + \gamma$
$^3_2\mathrm{He} + {}^3_2\mathrm{He} \longrightarrow {}^4_2\mathrm{He} + {}^1_1\mathrm{H}$
$^1_2\mathrm{H} + {}^1_1\mathrm{H} \rightarrow {}^2_1\mathrm{D} + {}^0_{+1}\mathrm{e}\ (\beta^+)$
$^2_1\mathrm{D} + {}^1_1\mathrm{H} \rightarrow {}^3_2\mathrm{He}$
$^3_2\mathrm{He} + {}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + {}^0_{+1}\mathrm{e}\ (\beta^+)$
$4\ {}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + 2\,{}^0_{+1}\mathrm{e}\ (\beta^+)$

Production of energy from fusion reactions?

The most promising fusion reaction on Earth is the combining of deuterium and tritium isotopes to produce a helium nucleus, a neutron and energy.

The mass difference is converted to energy according to Einstein's $E=mc^2$ equation.

$^2H + {}^3H \rightarrow {}^4He + n + 17.6$ MeV



(2) Fisson reaction

A nuclear fission reaction is a process where the nucleus of a heavy atom, such as Uranium-235, splits into two or more smaller nuclei after absorbing a neutron.

This splitting releases a large amount of energy more neutrons.

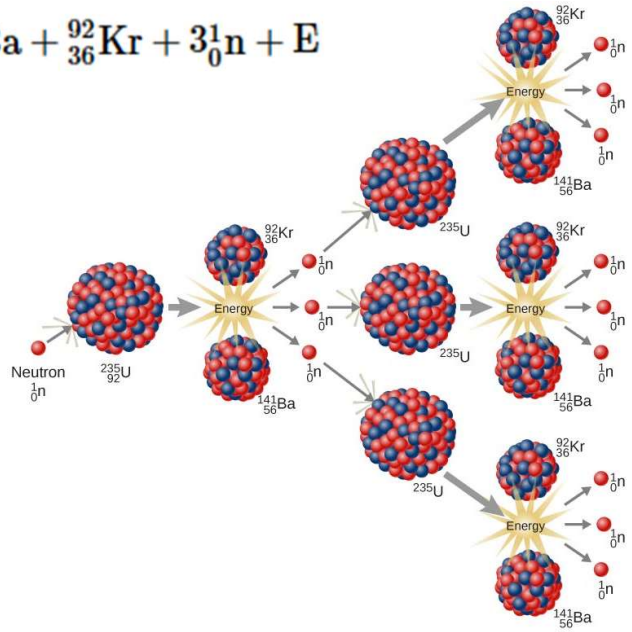
These fast neutrons produce more fissions creating chain reactions.

The Fission Process

- A neutron strikes the nucleus of a heavy, unstable atom (like U-235).
- The U nucleus absorbs the neutron and becomes extremely unstable (U*-236).
- The unstable nucleus splits into two or more lighter nuclei. (e.g. Ba and Kr).
- The fission process also releases a few more neutrons and an enormous amount of energy.

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{236}_{92}\mathrm{U} \rightarrow ^{141}_{56}\mathrm{Ba} + ^{92}_{36}\mathrm{Kr} + 3^{1}_{0}\mathrm{n} + \mathrm{E}$$

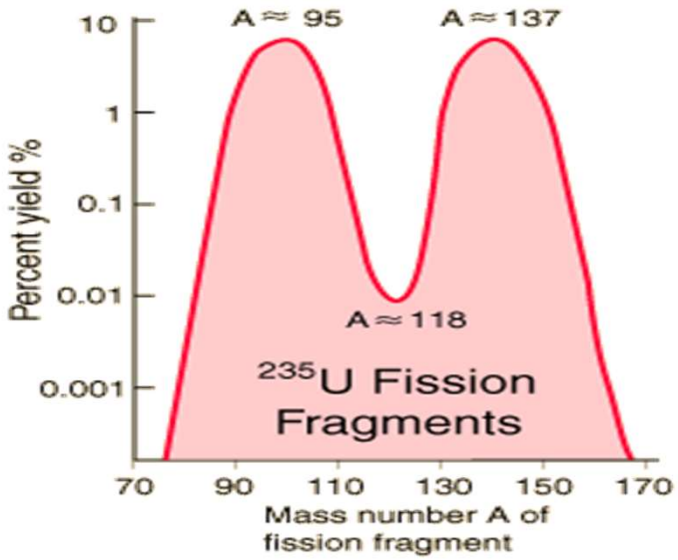
$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{236}_{92}\mathrm{U} \rightarrow ^{141}_{56}\mathrm{Ba} + ^{92}_{36}\mathrm{Kr} + 3^{1}_{0}\mathrm{n} + \mathrm{E}$$



When U*-236 undergoes fission, the average of the fragment mass is about 118.

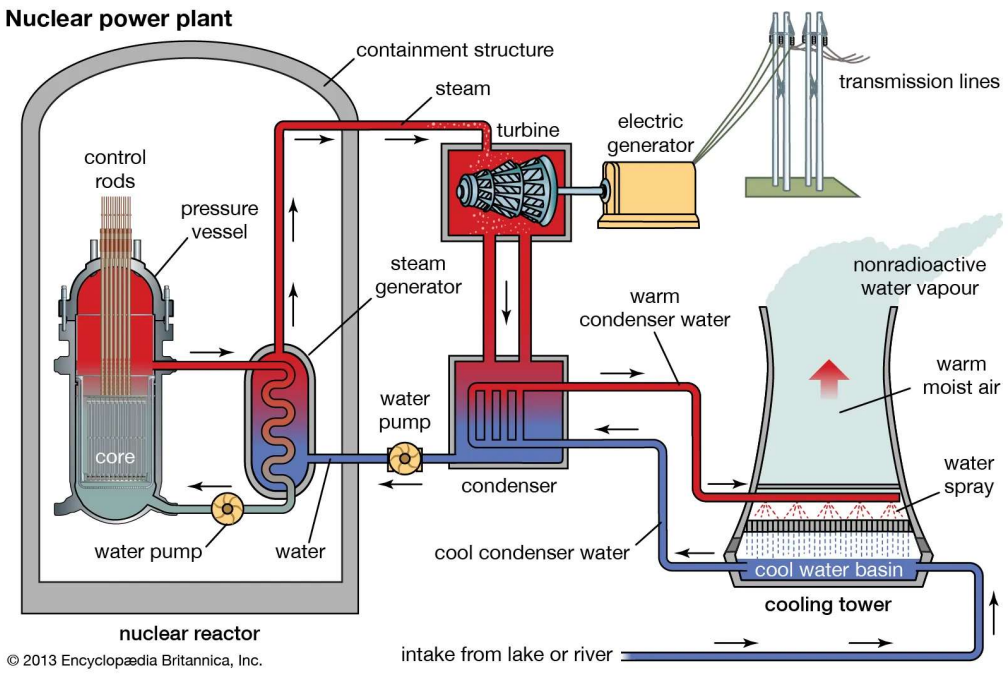
But very few fragments for that average are found.

Unequal fragments are most probable and their masses are around mass number 95 and 137.



Nuclear Reactors

- A free **slow neutron** strikes a fissile nucleus (such as uranium-235)
- Then the nucleus becomes **unstable and splits** into smaller nuclei, releasing a huge amount of heat energy and several (2-3) **fast-moving neutrons**.
- These new fast neutrons strike other fissile nuclei releasing more neutrons and, thus maintaining **a chain reaction**.
- **A moderator** (like heavy water or graphite) slows down the fast neutrons to thermal energy.
- **The control rods** made of neutron-absorbing materials, are inserted into the reactor core to absorb excess neutrons and control the rate of the reaction.
- The heat produced by the controlled fission is used to heat a water which turns **into steam**. This steam drives a **turbine to generate electricity**.



There are two types of reactors

1. Thermal Reactors

2. Fast Reactors

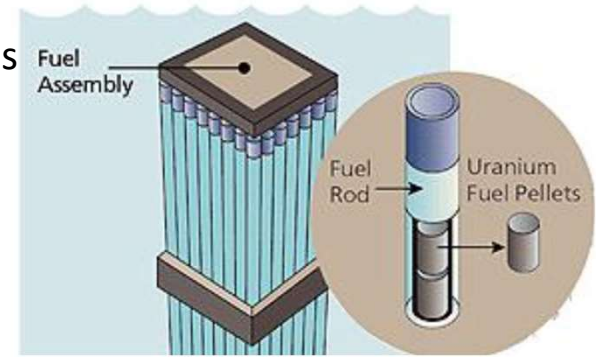
Basically there are two main parts in a thermal reactor.

- The **core** of the reactor which contains the **fuel, moderator, coolant,** etc.

- The **steam cycle** in which the heat produced in the core is used to produce steam in order to turn turbines.

The **core** of the reactor

The core contains
- fuel
- moderator
- coolant



In fast reactors there is no moderator.

Moderator

Present only in Thermal Reactors.

Used to slow down fast neutrons (about 2 MeV) to thermal energies.

Nuclei of low mass numbers are more effective.

Water, Heavy water and are often used.

Beryllium and BeO is also used occasionally but costly.

Coolant

- Coolant is used to remove heat from the core and from other parts of the reactor where heat may be produced.
- In Thermal Reactors Water, Heavy water and various gasses are used. In the case of water and heavy water, they are frequently serve as the moderator also.
- In Fast Reactors water and heavy water can not be used as coolant because they tend to slow down neutrons. Fast Reactors are cooled by liquid Sodium which has an excellent heat transfer properties. Gases can also be used in Fast Reactors.

Control Rods

Control rods are movable pieces of neutron absorbing material used to control the fission process in start-up, running process and shut down of the reactor. The movement of the rods in and out from the fuel assembly affects the multiplication factor *k* of the system.

Material Used:

- Boron Steel because Boron has high absorption cross section
- Hafnium of Cadmium.
- Silver or various alloys of Silver.

Nuclear Reactor Fuels

Nuclei such as ^{235}U that lead to fission, following the absorption of a zero energy neutron are called **fissile**.

Natural uranium has 0.7 wt% of ^{235}U and 99.3 wt% of ^{238}U
.

Despite this low concentration of ^{235}U it is possible to fuel certain types of critical reactors with natural uranium.

URANIUM

- Uranium is a fairly common element in the Earth's crust. Uranium is approximately as common as tin or germanium in Earth's crust, and is about 40 times more common than silver.
- Uranium is a constituent of most rocks, dirt, and of the oceans. The fact that uranium is so spread out is a problem because mining uranium is only economically feasible where there is a large concentration.
- The cost of nuclear power lies for the most part in the construction of the power station.



Advantages of Nuclear Energy

- **Clean energy**

  Nuclear energy is a clean energy source that doesn't produce smoke or harmful byproducts like fossil fuels

  It has a low carbon footprint, emitting minimal greenhouse gases

- **High energy density**

  Uranium has a high energy density, so it can be stored more easily

  Fission of one atom of uranium produces 10 million times more energy than burning one atom of carbon

- **Reliable energy**

  Nuclear energy is a reliable energy source that can keep the grid online

  It can power cities and towns more reliably than other energy sources

- **Combating climate change**

  Nuclear energy is a crucial component in combating climate change

  It can help reduce greenhouse gas emissions and mitigate global warming

- **Economic benefits**

  Nuclear energy can create thousands of jobs and add billions to the economy

- **Other benefits**

  Nuclear energy can protect air quality

  It can stay on no matter what Mother Nature throws at it

Disadvantages:
1. Nuclear Energy Raw Material

- Cost - Nuclear power is more expensive than other forms of electricity generation, such as wind, solar, and gas
- Uranium is used in the process of fission.
- While the uranium is being mined, transported and transformed into pellets there is a risk.
- This releases harmful radiation into its surroundings.
- Can be harmful to those handling the material.
- Runoff from the uranium mines contaminate ground water.

2. Water Pollution:

Nuclear fission chambers are cooled by water. This water is then turned into steam to power the turbines.

When the used water is cool, it is pumped into nearby wetlands. Then heavy metals and pollutants can leak.

The high heat energy of hot water can damage to eco systems near the reactor.

3. Nuclear Waste:

When the uranium is reacted radioactive byproducts are produced as waste.

The storage of the nuclear waste could contaminate through leaks or containment failures.

This waste could also become a target of terrorists seeking to use the waste to create nuclear weapons.

4. Shutdown of Reactors:

Nuclear reactors which fails or old must be shutdown.

These abandoned reactors are taking up valuable land space.

Could be contaminating the areas surrounding them

Also they are unstable to be removed.

5. Risk of accidents

Nuclear accidents can have catastrophic consequences, releasing radioactive materials and causing long-term environmental damage

Nuclear reactors are vulnerable to natural disasters, such as the Fukushima Daiichi accident

- **Other disadvantages**

  Nuclear power is a non-renewable fuel source

  Nuclear power plants consume a lot of water

  The installation cost of nuclear power plants is high

Other disadvantages

- Nuclear power is a non-renewable fuel source
- Nuclear power plants consume a lot of water
- The installation cost of nuclear power plants is high
- Nuclear power production leaves a large amount of dangerous radioactive waste.
- Nuclear waste must be sealed underground in protected chambers.

While nuclear plants are expensive to build but a single power plant can provide massive output for years.